

Chapter

Safety in Sports

1. Which fracture shows no break and no wound in the skin? SQP 2021

- (a) Linear fracture
- (b) Impacted fracture
- (c) Compound fracture
- (d) Simple fracture

2. What is a shin splint injury? MAIN 2024

- (a) It is a traumatic injury to the brain that alters mental status.
- (b) It is a partial or complete break in the continuity of the bone.
- (c) Pain in the lower region of the leg between the knee and ankle.
- (d) It is a car or sprain of the anterior cruciate ligament.

3. Lower back pain refers to what? COMP 2023

- (a) A tear or sprain of the medial collateral ligament.
- (b) Refers to any pain in the bottom region of the spine.
- (c) Occurs when the ligament that supports the ankle stretch beyond their limit.
- (d) In the lower leg between the knee and ankle.

4. Which of the following is a preventive measure for tennis elbow?

- (a) Avoiding all physical activity
- (b) Regular stretching and strengthening exercises
- (c) Keeping the arm completely still for long periods
- (d) Playing without warm-up

5. Which of the following is NOT a preventive measure for sports injuries? SQP 2024

- (a) Warming up properly before training
- (b) Wearing appropriate protective gear
- (c) Playing beyond physical limits
- (d) Ensuring the sports area is safe and clean

6. Which of the following is the primary cause of a sprained ankle? COMP 2021

- (a) Overstretching muscles in the back
- (b) Ligament injury from falling or twisting the ankle
- (c) Lifting heavy weights incorrectly
- (d) Sitting for long hours

7. Assertion (A) : ACL injuries commonly occur due to sudden stopping and pivoting movements.

Reason (R) : Strengthening exercises and maintaining alertness while playing can help prevent ACL injuries. **SQP 2021**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

8. Assertion (A) : Shin splints cause pain and inflammation in the lower leg.

Reason (R) : Using supportive footwear and regular strengthening exercises reduce the risk of shin splints. **COMP 2023**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

9. Assertion (A) : A sprained ankle occurs due to a ligament injury from twisting or falling.

Reason (R) : The PRICE method (Protect, Rest, Ice, Compression, Elevation) aids in recovery. **SQP 2022**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

10. Define first aid. MAIN 2024

11. What is meant by concussion? How can it be prevented? SQP 2021

12. Expand ACL and MCL. COMP 2023

13. Identify the sports injury given below.



- (a) Achilles Tendinitis
- (b) Shin Splints
- (c) Concussion
- (d) Sprained Ankle **SQP 2021**

14. Sneha, a professional basketball player, jumps for a rebound but lands awkwardly, twisting her knee inward. She experiences severe pain on the inner side of her knee and struggles to put weight on her leg. The team's physiotherapist suspects a torn medial collateral ligament (MCL) due to the sudden rotational stress on the knee. COMP 2023

- (i) What are the common causes of an MCL tear, and how does it affect knee stability?
- (ii) What preventive measures should athletes take to reduce the risk of an

MCL tear and what first aid protocol should be followed immediately after injury?

15. Rohan, a professional sprinter, has been training rigorously for an upcoming championship. During an intense sprinting session, he suddenly feels a sharp pain in his calf muscle, making it difficult for him to continue running. His coach examines the area and notices swelling and tenderness, indicating a possible muscle strain. Rohan recalls that he skipped his usual warm-up routine that day.

(i) What is a muscle strain, and what could have caused Rohan's injury?

(ii) What first aid steps should Rohan follow to recover from the strain and prevent further damage?

16. Mention any three sports-related injuries. MAIN 2024

Solution

1. (d) Simple fracture

A simple fracture results from excessive force but does not penetrate the skin.

2. (c) Pain in the lower region of the leg between the knee and ankle.

Shin splints refer to pain along the shinbone (tibia) caused by overuse or repetitive stress on the lower leg muscles and bone. They commonly occur in runners, dancers, and athletes who increase training intensity suddenly. It is not a fracture, brain injury, or ligament sprain.

3. (b) Refers to any pain in the bottom region of the spine.

Explanation: Lower back pain occurs in the lumbar region of the spine and is commonly caused by muscle strain, poor posture, or overexertion. It is not related to knee ligaments, ankle ligaments, or shin injuries.

4. (b) Regular stretching and strengthening exercises

Explanation: Tennis elbow is an overuse injury of the forearm muscles and tendons. Preventive measures include strengthening exercises, stretching, using proper technique, and allowing adequate rest to reduce strain on the tendons.

5. (c) Playing beyond physical limits

Overexerting the body can cause muscle fatigue, poor coordination, and a higher risk of injuries. Preventive measures include proper warm-ups, wearing suitable protective gear, and ensuring a safe playing environment.

6. (b) Ligament injury from falling or twisting the ankle

A sprained ankle happens when the ligaments supporting the ankle are stretched or partially torn, usually due to sudden twisting, rolling, or falling on an uneven surface. Proper rest, ice, compression, and elevation (RICE) help in recovery.

7. (a) Both A and R are true, and R is the correct explanation of A.

ACL (Anterior Cruciate Ligament) injuries often occur during sudden stops, pivots, or changes in direction, common in sports like football and basketball. Strengthening exercises enhance knee stability, and being alert during play reduces the risk of injury, making the reason correctly explain the assertion.

8. (a) Both A and R are true, and R is the correct explanation of A.

Shin splints cause pain and inflammation in the lower leg due to repetitive stress on the shinbone and surrounding muscles. Using supportive footwear and performing strengthening exercises help reduce this stress, preventing the injury.

9. (a) Both A and R are true, and R is the correct explanation of A.

A sprained ankle happens when the supporting ligaments are overstretched or torn due to twisting, rolling, or falling. The PRICE method—Protect, Rest, Ice, Compression, Elevation—helps reduce swelling, alleviate pain, and stabilize the injured area, promoting proper recovery.

10. First aid is the immediate and temporary care given to an injured or sick person until professional medical help can be obtained. It aims to preserve life, prevent the condition from worsening, and promote recovery.

11. A concussion is a type of brain injury that occurs when the head experiences a sudden impact or blow, causing the brain to move rapidly inside the skull. It can be prevented by wearing proper protective headgear, such as helmets, during sports, cycling, or other activities that pose a risk of head injury.

12. ACL : Anterior Cruciate Ligament

MCL : Medial Collateral Ligament

13. (b) Shin Splints

Shin splints cause pain and inflammation in the front of the lower leg, commonly occurring in runners and athletes engaged in repetitive, high-impact activities.

14. (i) Causes and Effect of MCL Tear:

An MCL (Medial Collateral Ligament) tear commonly occurs due to excessive twisting of the knee, sudden changes in direction, or a direct blow to the outer side of the knee that forces it inward. This injury compromises knee stability, making it difficult to resist inward forces, walk, or perform lateral movements, and can cause pain and swelling on the inner side of the knee.

(ii) Preventive Measures and First Aid:

Prevention:

- Strengthen muscles around the knee through conditioning exercises.
- Avoid sudden lateral movements or excessive twisting.
- Use knee braces or supports during high-risk activities.

First Aid (Immediately After Injury):

- **RICE Protocol:** Rest, Ice, Compression, Elevation to reduce swelling and pain.
- **Immobilization:** Avoid putting weight on the injured leg.
- **Pain Management:** Take mild painkillers if necessary.
- **Medical Consultation:** Seek professional evaluation to determine severity and treatment.

15. (i) Muscle Strain and Cause:

A muscle strain occurs when a muscle or its tendon is overstretched or partially torn due to sudden force, overuse, or improper movement. In Rohan's case, skipping his warm-up, performing a high-intensity sprint, and sudden forceful contraction of the calf muscle likely caused the strain.

(ii) First Aid Steps for Recovery and Prevention:

1. **Rest:** Avoid using the injured leg and stop sprinting or strenuous activity.
2. **Ice Application:** Apply ice wrapped in a cloth for 15–30 minutes to reduce pain and swelling.

3. **Compression:** Use an elastic bandage to stabilize the muscle and limit movement.
4. **Elevation:** Keep the leg raised above heart level to minimize swelling.
5. **Pain Relief:** Take mild painkillers (e.g., paracetamol) if necessary.
6. **Prevention:** Always perform proper warm-up and stretching exercises before intense training sessions, and gradually increase intensity.

16. Three common sports-related injuries are:

1. **Muscle Strains** – Overstretching or tearing of muscle fibers, causing pain and limited movement.
2. **Shin Splints** – Pain in the lower leg due to stress or overuse, often seen in runners.
3. **Bone Injuries** – Fractures or cracks caused by impact, overuse, or falls during sports activities.